

Rachana Nijalingappa

• nrachananijalingappa@gmail.com • (+44)7768923731 • [LinkedIn](#) • London, UK • [GitHub](#)

PROFILE SUMMARY

AI-driven Software Engineer with over 5+ years of experience in architecting scalable, cloud-native systems across **FinTech and Healthcare sectors**. Proven track record in transitioning legacy architectures to **Python-based** automated workflows and deploying high-security backend solutions using **C#, .NET and Azure**. Currently pursuing an **MSc in Artificial Intelligence** to leverage expertise in **Agentic AI, LLMs, NLP, and Deep Learning** to solve complex enterprise problems.

WORK EXPERIENCE

SOFTWARE ENGINEER

Dec 2023 – Jan 2025

Commonwealth Bank of Australia, India, Bangalore

- Engineered secure backend APIs in .NET 8.0 for the Tokenization project, enhancing transaction privacy and securing sensitive data using Azure Key Vault and Storage.
- Spearheaded the migration of 21 observability dashboards and log streams from Splunk to Observe, optimizing system performance tracking and monitoring efficiency.
- Delivered high-reliability features by managing full-cycle sprints and achieving 80% test coverage via NUnit, significantly reducing post-production defects.
- Tech Stack:** .NET 8.0, C#, Azure (Storage, Key Vault, Service Bus), NUnit, Observe, Splunk, SQL.

ASSOCIATE TECHNOLOGY L2

May 2022 – Dec 2023

Publicis Sapient, India, Bangalore

- Led the technical migration of a banking project from .NET Core to Python, enabling advanced data scraping, cleaning, and automated backend workflows.
- Optimized data quality and visualization by developing automated processing scripts and integrating Python backend data with Angular dashboards.
- Streamlined release cycles by architecting CI/CD pipelines in GitHub and managing daily WebJobs to ensure consistent availability of critical banking data.
- Tech Stack:** .NET 6.0, Python (Flask), Azure (Functions, Service Bus), SQL, Angular, Docker, NUnit, NumPy, Pandas.

SOFTWARE ENGINEER

July 2019 – May 2022

Harman, India, Bangalore

- Developed full-stack IoT and Healthcare applications for NHS and Legrand, building UI components in Angular to improve patient data accessibility.
- Secured healthcare and IoT data by implementing JWT-based authentication and designing scalable cloud integrations with Azure API Gateway and Storage.
- Engineered real-time event handling using Azure Functions and Service Bus for asynchronous communication, while maintaining 75% test coverage with NUnit and Moq.
- Tech Stack:** .NET Core, SQL, Angular, Azure (Functions, Service Bus, Application Insights, API Gateway), NUnit, Moq.

PROJECTS

Household Sustainability Data Analysis

- Conducted advanced statistical analysis using Stepwise AIC selection to optimize a linear model explaining 86.6% of the variance in household carbon footprints.
- Applied Binary Logistic Regression to evaluate the impact of socio-economic factors and environmental concern on recycling habits.
- Utilized R Markdown to create a fully reproducible research report, ensuring transparency in data cleaning, model validation, and diagnostic testing.

Tech Stack: R (MASS, car, lmtest), R Markdown

ShopNova - E-Commerce Modular Monolith

- Designed a modular monolith with 3 schema-separated bounded contexts and 13 RESTful API endpoints.
- Implemented CQRS with MediatR, cross-module domain events, and automated Fluent Validation pipeline.
- Built observability layer with Serilog structured logging and per-request correlation ID tracing.

Tech Stack: .NET 8, EF Core, MediatR, JWT, Ocelot

Credit Card Fraud Detection

- Built an end-to-end ML pipeline on 283,726 real transactions (0.17% fraud rate), training and comparing 13 models across classical ML, gradient boosting, and deep learning (LSTM, 1D-CNN, Autoencoder)
- Applied SMOTE exclusively on training data to address class imbalance, preventing data leakage and improving fraud recall to 76.8%.
- Achieved 85.9% F1-Score and 97.3% precision using Optuna-tuned LightGBM (30-trial Bayesian search), best among all 13 models evaluated.
- Implemented SHAP explainability to identify top fraud-predictive features, and unsupervised anomaly detection using Isolation Forest and K-Means clustering.

Tech Stack: Python, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Jupyter.

AREA OF EXPERTISE

Languages:	C#, Python, R, JavaScript, HTML, CSS, .NET, NumPy, Pandas, Scikit-learn.
AI & ML:	Generative AI, LLMs (GPT, OpenAI, Claude, Gemini), Deep Learning, NLP, Neural Networks, ML Algorithms, RAG, TensorFlow, PyTorch.
Data Visualization:	Data Cleaning, Statistical Analysis, Predictive Modeling, Matplotlib, Seaborn, Plotly.
Cloud & DevOps:	Microsoft Azure (Functions, Service Bus, Key Vault, App Insights), AWS, Docker, Kubernetes, CI/CD.
Backend & APIs:	RESTful APIs, Microservices, MVC Architecture, CQRS, Entity Framework, LINQ.
Databases:	SQL, Azure Storage, MongoDB, Excel.
Testing & Security:	NUnit, Integration Testing, JWT Authentication.
Tools:	GitHub, Visual Studio, VS Code, Jupyter, Bitbucket, JIRA, Postman, Observe, Swagger.
Methodologies:	Agile/Scrum, SOLID Principles, Dependency injection.

CERTIFICATES

- Databricks Accredited AI Agent Fundamentals
- Forage – Risk Job Simulation and Data Science Job Simulation
- Anthropic Certificate of Completion: AI Fluency Framework & Foundations
- Certificate of Training - Cisco AI Fundamentals
- Certificate of Training - Introduction to Prompt Engineering for Generative AI

EDUCATION

BRUNEL UNIVERSITY

MSc Artificial Intelligence

London, UK

2025-2026

MVJ COLLEGE OF ENGINEERING

Bachelor of Electronics and Communication (BE) [CGPA: 8.28/10]

Bangalore, India

2015-2019